School of Computer & Systems Sciences
Jawaharlal Nehru University
CS-429, Cyber Security
MCA III – Semester
Mid Semester Examination, 2023

Time: 90 minutes **Max. Marks: 25**

Note: Attempt any five questions. Each Question 5 marks

1. Explain CIA Triad with an example of each.

2. Let ciphertext is DWWDFNDWGDZQ. Find the key using Brute Force of simple substitution cipher.

3. Decrypt the given ciphertext WWVA using the Hill cipher with a key DCDF.

4. Using the Vigenère cipher, encrypt the word "explanation" using the key leg

5. A ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is "B," and the second most frequent letter of the ciphertext is "U". What are the values of **a** and ***b*** to break an affine cipher code.

6. Alice uses the same key k to encrypt two messages m_1 and m_2 to get ciphertexts $c_i = E(k, m_i) = k \oplus m_i$ using One Time Pad. Eve later manages to learn the message m_2 in addition to both ciphertexts c_1 and c_2. Show how Eve can reconstruct m_1 with the available information.

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MCA III – Semester
End Semester Examination, 2023

Time: 3 hours **Max. Marks: 70**

Note: Attempt any 7 questions. Each question is of 10 marks.

1. Encrypting the message "**Hide the gold in the tree stump**" using Playfair cipher with the Key: '**PLAYFAIR EXAMPLE**'.

2. Decrypt the message "**dytyvcyynabvn**" by applying Vigenère Cipher with the key '**cute**'.

3. Suppose that the following is an excerpt from the decryption codebook for a classic codebook cipher.

123	once
199	or
202	maybe
221	twice
233	time
332	upon
451	a

   Decrypt the following ciphertext: 242, 554, 650, 464, 532, 749, 567 assuming that the following additive sequence was used to encrypt the message 119, 222, 199, 231, 333, 547, 346

4. For bits x, y, and z, the function maj(x, y, z) is defined to be the majority vote, that is, if two or more of the three bits are 0, then the function returns 0; otherwise, it returns 1. Write the truth table for this function and derive the boolean function that is equivalent to maj(x,y,z).

5. This problem deals with the DES cipher.
   a. How many bits in each plaintext block?
   b. How many bits in each ciphertext block?
   c. How many bits in the key?
   d. How many bits in each subkey?
   e. How many rounds?
   f. How many S-boxes are used in each round?
   g. An S-box requires how many bits of input?
   h. An S-box generates how many bits of output?

PTO

6. This problem provides a numerical example of encryption using AES.

a. Given the current state matrix as $\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$. Show the value of State after ShiftRows.

b. Let the current state matrix as $\begin{pmatrix} 00 & 3C & 6E & 47 \\ 1F & 4E & 22 & 74 \\ 0E & 08 & 1B & 31 \\ 54 & 59 & 0B & 1A \end{pmatrix}$. Show the value of State after SubBytes. The S-Box is given below.

S-Box

AES S-Box: Substitution values in hexadecimal notation for input byte (*xy*)

x \ *y*	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

7. Perform encryption and decryption using RSA algorithm for the following:
   a. $p = 3$, $q = 11$, $d = 7$, Encrypt the message $M = 5$.
   b. $p = 5$, $q = 11$, $e = 3$, Decrypt the ciphertext $C = 14$.

8. How does the Diffie Hellman key exchange work? What is the weakness of Diffie Hellman key exchange?

9. What is digital signature? What are the importance of it? Explain the procedure to create a digital signature with a diagram.